

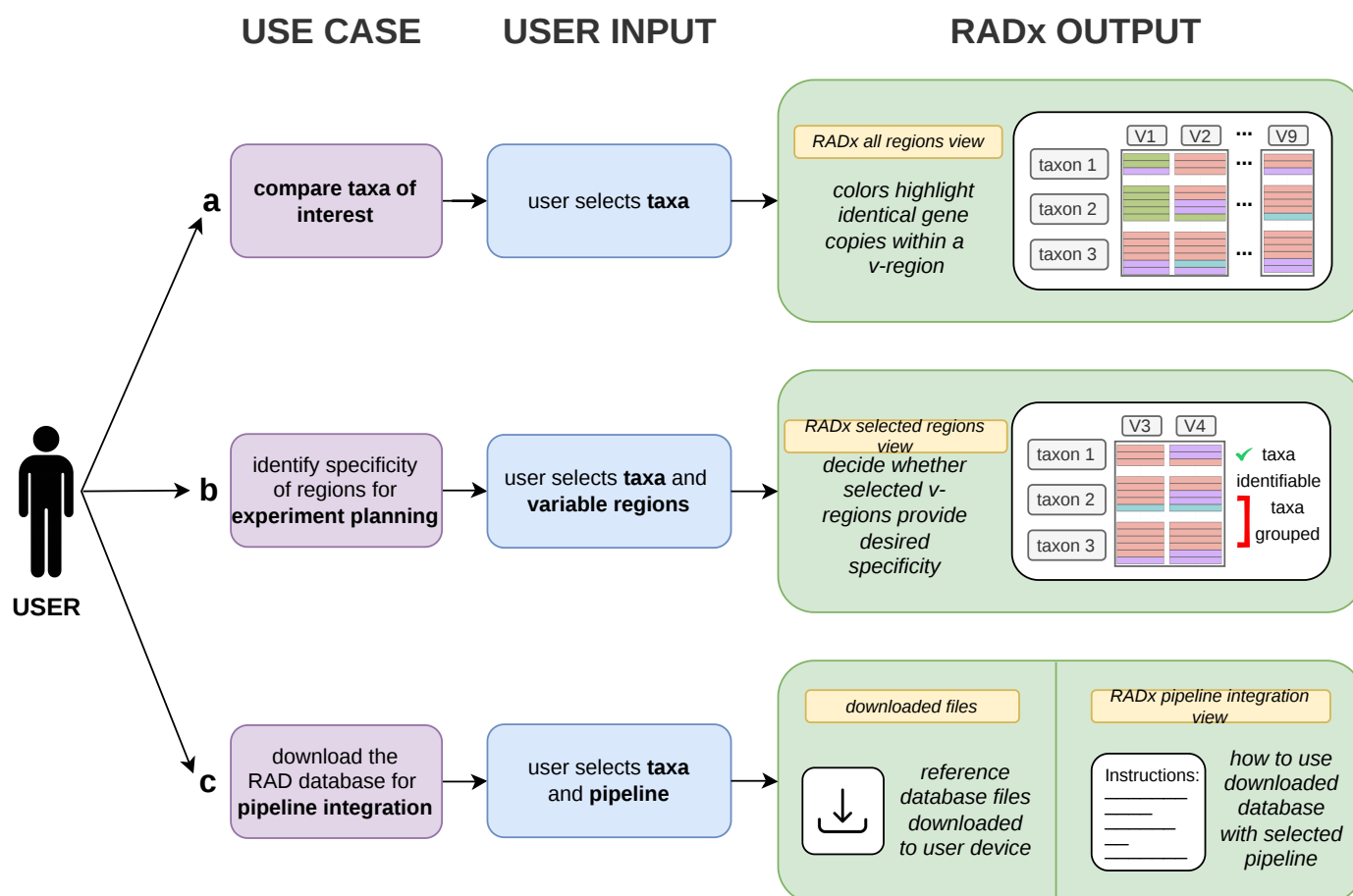


Figure 1: Use cases for RADexplorer. (a) To compare taxa of interest, the user selects the taxa from a list of all genera and species included in the RADlib sequence library. The RADalign package filters the selected taxa out of RADlib, performs a multiple sequence alignment on each variable region (v-region), and returns the results of the alignment to the RADexplorer app (RADx). In RADx's all regions view, each gene copy present in a taxon is displayed as a colored bar. Bars with the same color indicate sequences that align exactly within each v-region. (b) Once selected taxa have been displayed with all v-regions, the user can select specific v-regions to filter the display. RADx will also display which taxa are uniquely identifiable with the selected v-regions and which taxa will be grouped due to sharing v-region sequences. (c) The user can also use RADx to integrate RADlib with existing 16S analysis pipelines (currently supports MetaScope). RADalign downloads either the full RADlib or a user-selected set of sequences from RADlib, and displays instructions for how to use the downloaded files as the reference database for a given pipeline.